



Open-source AI 101: the battle for the future of AI

July 21, 2026

Open-source versus proprietary AI models in a nutshell

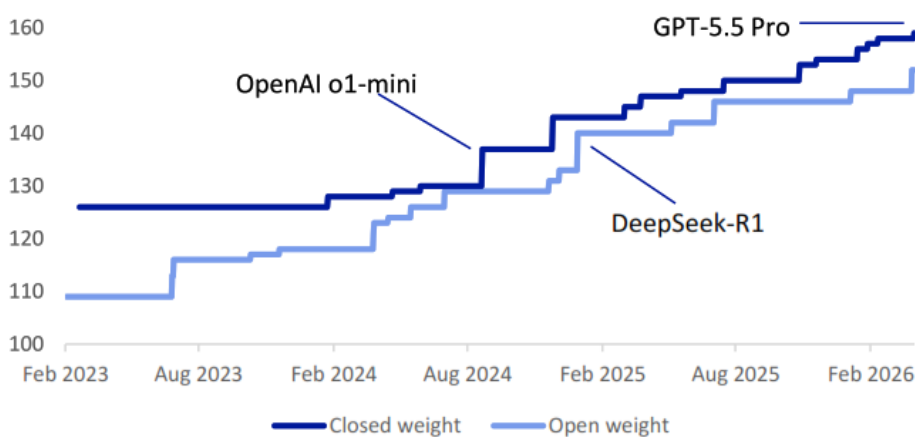
AI-related stocks tumbled at the end of last week on a second DeepSeek moment, as the latest Chinese open-weight AI model reminded investors that cheap, light alternatives are snapping at the heels of the state-of-the-art proprietary models.

In this brief report for generalists, we explain this latest technology format war:

1. Open-what? A new twist on age-old format wars
2. What are open AI models?
3. Why are they disrupting the markets?
4. Why would a company give its model away?
5. What comes next?

The emergence of DeepSeek's breakthrough R1 model in January last year wiped \$600bn off the market value of Nvidia in one day on concerns about whether vast hyperscaler capital expenditure was justified in the light of a "good enough" model developed in China using older chips and trained for a fraction of the cost.

Figure 1: Epoch Capabilities Index score: open-weight models narrow the gap with closed-weight proprietary models (data to end April 2026)



Source : Epoch AI, Deutsche Bank Research

Authors

Adrian Cox
Research Analyst



A similar release on Friday revived those concerns, with the Magnificent Seven (mostly hyperscaler) stocks shedding 1.8 percent and the Philadelphia Stock Exchange Semiconductor Index turning down 1.6 percent, bringing its loss for the week to more than 10 percent. Alphabet's Google, Microsoft, Amazon's AWS, Meta and Oracle are set to spend about \$700bn building AI capacity this year, up by about 70 percent from last year.

Even so, while powerful open models raise questions about the business models of proprietary model makers and semiconductor demand, they may be bullish for AI adoption, applications and enterprise customisation. Greater competition, more efficient models and lower prices may supercharge a new phase of growth.

1. Open-what? A new twist on age-old format wars

Open-source software is nothing new.

- A [study](#) by Harvard Business School in 2024 found that about 96 percent of commercial programs include some code created or distributed for free by public-facing tech forums. Without it, enterprises would pay 3.5 times more, or roughly \$8.8 trillion, to build their software and platforms, it found.

But open AI models are contentious.

- Some critics saying that once released they may be less secure and more likely to be misused. (We mainly use the term "open models" for readability and explain the distinction between open-source and open-weight in the definition section.)
- Some open models have also been [accused](#) of being trained through "distillation", using the outputs of more capable proprietary models to teach a new model, which may breach providers' terms and magnify errors, biases or safety weaknesses in the original models. Critics also wonder out loud about whether some open models have been geared more towards scoring highly on benchmarks – so-called [benchmaxxing](#) – than for solving real-world problems.

Meanwhile, concerns have been rising about whether proprietary models can be [depended on](#).

- Anthropic's most powerful Claude models were recently de facto banned for several weeks on security concerns. Critics also say calls by proprietary model makers for safety regulation would in fact protect incumbents.

The debate is reminiscent of past format battles.

- Every great technology needs a rival. In the "war of the currents" of the late 1880s, Thomas Edison defended his direct-current (DC) empire by portraying the alternating-current (AC) system promoted by George Westinghouse and Nikola Tesla as lethally dangerous—even supporting demonstrations in which animals were electrocuted with AC.
- And remember VHS versus Betamax video cassette players in the late 1970s and early 1980s? VHS did not win by being the better technology: JVC licensed it broadly, manufacturers produced cheaper machines, recording times were longer and a self-reinforcing ecosystem developed around it. Open-source AI could follow the same path. It does not need to beat every proprietary model on every benchmark. It needs to be good enough, inexpensive and ubiquitous.



However, while the contest between open and proprietary AI is often presented as a winner-takes-all format war, the most likely outcome is a combination of the two.

- At the consumer level, it may resemble Apple and Google's Android, a controlled premium ecosystem coexisting with a larger, more customisable ecosystem. And at the industry level, it may resemble the IBM personal computer, where widespread compatibility commoditised the foundational technology while shifting significant value into chips, software, services and applications built around it. Many organisations will use both.

2. What are open AI models?

There are three distribution models. For users, it boils down to control and – to coin a phrase – with greater control comes greater responsibility.

Proprietary, or closed, models are generally the plug and play option, accessed through the developer's application, an application programming interface (API), or a managed cloud services.

- For a rough analogy, think of Apple's iOS operating system or Microsoft's Windows, where the company controls the underlying technology and the terms on which others can use it.
- With proprietary AI models, the developer keeps the model weights, training methods and most details of the training process private. It controls pricing, updates and safety restrictions, and can alter or withdraw access. Prominent examples include OpenAI's flagship GPT models and Anthropic's Claude models.

Open-weight models are open-source-lite.

- Their trained parameters – the billions or trillions of numerical values that encode much of the model's learnt behaviour – are typically released for users to download, subject to an applicable licence. You can run the model on your own infrastructure, adapt or fine-tune it, and technically alter or remove its safeguards.
- But the release may not include the original training data, the complete training code or enough information for you to reproduce the model from scratch. Many of the models referred to casually as open-source – including DeepSeek's R1 and Meta's Llama models, as well as many released by France's Mistral – are in fact more accurately described as open-weight.

Genuinely open-source AI systems are much rarer.

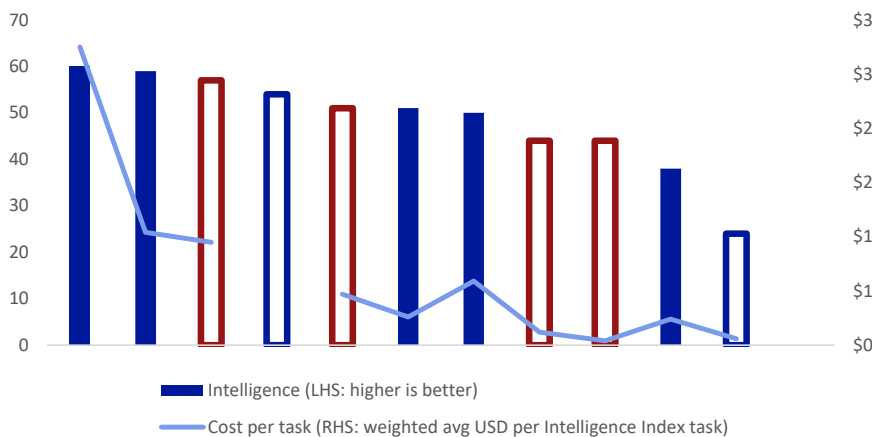
- Under the [Open Source Initiative](#), they should give users the freedom to use, study, modify and redistribute the system, supported by access to its parameters, training and inference code and detailed information about the training data. The grand-daddy of open-source is Linux, a free operating system created in 1991 and found in nearly all cloud infrastructure and the majority of the world's servers and mobile devices.



Consequently, open-source models often offer:

1. **Lower costs, particularly at scale or for specialised tasks.** If you run the model yourself, you can avoid the developer's per-token API charges and can select smaller or compressed models requiring fewer calculations. **The downside?** Downloadable does not mean free: you still have to pay for compute, electricity, engineering, monitoring and maintenance. Open models may also not be as universally capable as the leading proprietary systems and may lack their polished integration with wider products and services.
2. **Greater customisation and control.** You can fine-tune a model for your industry, language or task, rather than relying on a one-size-fits-all service. **The downside?** Although open models are cheaper over the long-run, you need to invest a lot upfront in infrastructure and expertise. You have to take more responsibility for testing, guardrails, cybersecurity, updates and reliability—though managed cloud providers can fill some of the gaps.
3. **Local deployment.** You can install some models on your own servers, or even on "edge devices" such as cars and phones, rather than relying on a connection to the cloud network run by the provider. That makes them potentially faster (less "latency") and more secure as you don't need to share your data. Once you download it, it's yours, and API access cannot be suspended without warning. That is key for some regulated organisations where data can't leave the premises. **The downside?** Not every model can run economically on local hardware, and responsibility for maintaining performance and protecting the system falls on your shoulders.

Figure 2: Intelligence index of selected AI models vs cost per tasks: US (blue) and proprietary (solid colour) models are no longer far ahead of Chinese (red) and open-weight (white with border) models



Source : Artificial Analysis Intelligence Index, Deutsche Bank Research. Data as of July 20, 2026



3. Why are they disrupting the markets?

Open models challenge four assumptions about proprietary models embedded in AI valuations.

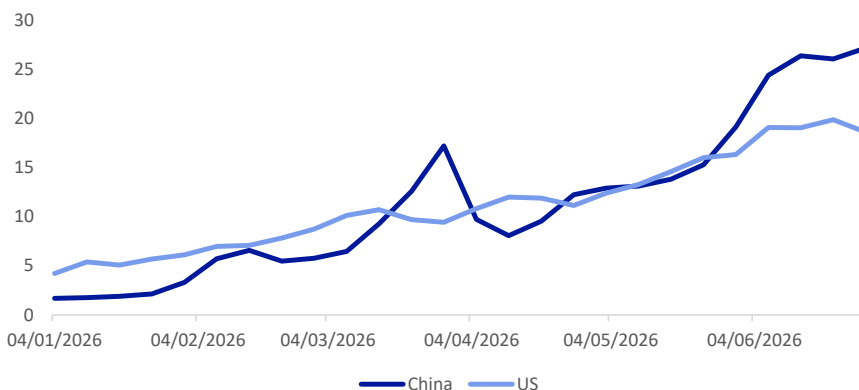
First, scarcity. If capable models can be copied, modified and hosted by many providers, AI models might begin to look like a commoditised software infrastructure rather than a uniquely monetisable product.

Secondly, pricing power. AI agents and “tokenmaxxing” employees are guzzling up vast numbers of tokens, the small units of data that AI models process. As pilots scale up and AI subsidies come to an end, the leading makers of comprehensive models are starting to charge for each token rather than a flat rate. For cost-conscious companies, open models that are good enough for most purposes are looking increasingly attractive.

Thirdly, capital investment. If you can get something like the same performance out of an open model with lighter training and inference (running) horsepower, can the projected spending on chips, networking equipment, electricity and data centres be justified?

Fourth, US technological moat. Chinese open models weaken the perceived durability of US supremacy. The Trump administration says it is in “a [race](#) to achieve global dominance in AI” and has driven policies to strengthen domestic production and adoption and stifle Chinese access to cutting-edge chips and other technology. Chinese models have overtaken US models this year in terms of the number of tokens processed on each by the OpenRouter developer platform.

Figure 3: OpenRouter traffic: trillion tokens going through model platform this year, broken down by country of model



Source : OpenRouter, Deutsche Bank Research

For more, see our recent report “[AI IPOs 101: why they’ll change the boom forever](#)”.



4. Why would a company give its model away?

Open models may be commercially viable because producers do not need to monetise the weights themselves. It's about losing the battle to win the war.

1. Monetising complementary products

- Developers can charge for convenient hosted APIs, premium subscriptions, faster inference, enterprise support, fine-tuning, security and guaranteed service levels. The equivalent of in-app purchases for a free app gives less tech-engaged customers the best of both worlds.

2. Building an ecosystem and standard

- A widely adopted model attracts developers, tools, optimisations, fine-tuned variants and research. All of that can improve the model, lower the cost of acquiring new customers, and make its interface an industry standard.

3. Commoditising competitors' profit pool

- This is a classic case of [commoditising the complement](#): giving away something for free (or very little) to stop your rival from earning excess returns, while increasing demand for (and monetising) your own surrounding infrastructure. So cloud providers that produce or encourage open models can benefit from selling compute to host open models; consumer platforms can benefit from better recommendations and advertising; and application companies can benefit from lower input costs.

4. Boosting reputation, talent and capital

- In a world where you could fit all of the elite AI researchers (about [5,000](#)) into a mid-sized concert hall, and where companies are racing to attract capital to fund exponentially increasing compute costs, a leading open model demonstrates technical competence, attracts researchers and polishes the case for fundraising or an initial public offering.

5. Increasing geopolitical influence

- Open models can spread a country's technical standards, software ecosystem and cultural standards internationally. For China, capable open models reduce its dependence on US APIs and present it as the provider of choice to emerging markets. France has championed Mistral's open models as a European alternative. The US is not sitting idle either. OpenAI itself released smaller, open-weight gpt-oss models as a defensive move last August, and Thinking Machines launched Inkling, its own open model, last week.



5. What comes next?

History shows that formats can coexist where each offers something different.

- The contest between doesn't have to be an either/or. With so much at stake, OpenAI's honeymoon of a near-monopoly on generative AI in the 18 months or so after the launch of ChatGPT in November 2022 was never going to last.
- This is in line with the oft-quoted Jevons Paradox, where more efficient steam engines increased, not decreased, demand for steam power – and hence coal. Similarly, demand for compute will likely continue to surge as the market bifurcates into a handful of cutting-edge proprietary models on the one hand and a myriad of smaller open-source models on the other – just like trains and cars both found their market and continue to live side by side. No one ever said: "That's enough intelligence for now".

But both will exert a positive gravitational force on each other.

- Open models will force proprietary models to keep a lid on prices and remain flexible; proprietary models will set the benchmark for industrial-grade quality, safety and fully integrated products. The competitive moat will increasingly move away from raw weights and towards the balance of proprietary data, product design, memory, security and efficiency.

The greatest unknown in this uneasy coexistence is the direction of AI regulation.

- A growing backlash against AI in the US and the West significantly raises the chance of dramatic policy that could significantly affect its trajectory. While a handful of key US model makers presents a large target for legislators, they are also able to speak with one voice in a way that is not possible for fragmented, often foreign, open-source providers.
- For example, Anthropic released an [Advanced AI Framework](#) last month that called for obligations on frontier AI developers to test their models for catastrophic risks, engage external evaluators and disclose incidents on an ongoing basis. Their message echoes warnings from scientists and economists ranging from job losses to extinction.
- But critics say onerous regulations could favour incumbents with deep enough pockets to cover the costs and would stifle innovation, putting national security at risk. Former White House AI tsar David Sacks posted over the weekend on social media that there was "no reason to limit American models on tasks that Chinese models handle without issue. We're only making ourselves less competitive".



Recent relevant reports

[AI IPOs 101: why they'll change the boom forever](#) (July 1, 2026)

[Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh](#) (June 9, 2026)

[ChatIPO: Quick take on OpenAI's record-breaking IPO plans](#) (May 21, 2026)

[Will Claude's AI finance agents change your life?](#) (May 6, 2026)

[AI, Iran and the Gulf 101](#) (April 13, 2026)

[What AI says about AI eating itself and the world...](#) (Feb 16, 2026)

[AI Q&A: Why have market vibes suddenly shifted on AI?](#) (Feb 16, 2026)

[Three AI themes for 2026: the honeymoon is over](#) (Jan 20, 2026)



Appendix 1

This material has been prepared by the Deutsche Bank Research Institute and is provided to you for general information purposes only. The Institute leverages the views, opinions, and research from Deutsche Bank Research, economists, strategists, and research analysts. Accordingly, you should assume that content in this document is based on or was previously published and provided to Deutsche Bank clients who may have already traded on the basis of it.

Any views or estimates expressed in this material reflect the current views of the author(s) and may differ from the views and estimates of Deutsche Bank AG, its affiliates, other Deutsche Bank personnel, and other materials published by Deutsche Bank. The content in this material is valid as of the date shown on the first page and may change without notice. Deutsche Bank has no obligation to provide any updates or changes to the information herein.

This material should not be used as a basis for trading securities or other financial products and should not be considered to be a recommendation or individual investment advice for any particular person. It does not constitute an offer, solicitation, or an invitation by or on behalf of Deutsche Bank to any person to buy or sell any security or financial instrument. Nothing in this material constitutes investment, legal, accounting or tax advice. Deutsche Bank engages in securities transactions, including on a proprietary basis, and may do so in a manner inconsistent with the views or information expressed in this material.

While information in this material has been obtained from sources believed to be reliable, neither Deutsche Bank AG nor any of its affiliates makes any representation or warranty, express or implied, as to the accuracy or completeness of the statements or any information contained in this material and therefore any liability is expressly disclaimed. This material is provided without any obligation, whether contractual or otherwise. Information regarding past transactions or performance is not indicative of future results.

In the US this report is approved and/or distributed by Deutsche Bank Securities Inc., a member of FINRA. In Germany this information is approved and/or communicated by Deutsche Bank AG Frankfurt, licensed to carry on banking business and to provide financial services under the supervision of the European Central Bank (ECB) and the German Federal Financial Supervisory Authority (BaFin). In the United Kingdom this information is approved and/or communicated by Deutsche Bank AG, London Branch, a member of the London Stock Exchange, authorized by UK's Prudential Regulation Authority (PRA) and subject to limited regulation by the UK's Financial Conduct Authority (FCA) (under number 150018) and by the PRA. This information is distributed in Hong Kong by Deutsche Bank AG, Hong Kong Branch, in Korea by Deutsche Securities Korea Co. and in Singapore by Deutsche Bank AG, Singapore Branch. In Japan this information is approved and/or distributed by Deutsche Securities Limited, Tokyo Branch. In Australia, retail clients should obtain a copy of a Product Disclosure Statement (PDS) relating to any financial product referred to in this report and consider the PDS before making any decision about whether to acquire the product.

By accessing this material, you agree that its content may not be reproduced, distributed or published by any person for any purpose, in whole or part, without Deutsche Bank's prior written consent. You also agree that you shall not scrape, extract, download, upload, sell or offer for sale any of the content in this material, and you agree not to use, or cause to be used, any computerized or other manual or automated program or mechanism, tool, or process, including any scraper or spider robot, to access, extract, download, scrape, data mine, display, transmit, or publish, any of the content in this material.